

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.Tech AI &DS	Date:	30-07-2026

Code of Conduct

I N.Kishan kumar-192424404 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

1. Inflectional Morphology and Porter Stemmer

Input Words

- Talked
- Talking
- Talks

Step 1: Morphological Breakdown

Word	Root	Suffix
Talked	Talk	-ed
Talking	Talk	-ing
Talks	Talk	-s

Step 2: Porter Stemmer

- Remove **-ed** → Talk
- Remove **-ing** → Talk
- Remove **-s** → Talk

Step 3: Classification

- Talked → Inflectional
- Talking → Inflectional
- Talks → Inflectional

Normalized Root

Talk

Role of Morphological Normalization

- Removes grammatical suffixes.
- Converts all word forms into one base word.
- Improves search accuracy.
- Reduces duplicate indexing.
- Makes information retrieval faster.

2. Hybrid Morphological Processing Pipeline

Input Words

- Relational
- Conditional
- Happiness
- Running

Step 1: Morphological Breakdown

Word	Root	Affix
Relational	Relate	-ational
Conditional	Condition	-al
Happiness	Happy	-ness
Running	Run	-ing

Step 2: Finite-State Parsing

Start → Root → Add Suffix → Final Word

- Relate → +ational → Relational
- Condition → +al → Conditional
- Happy → +ness → Happiness
- Run → +ing → Running

Step 3: Porter Stemmer

- Relational → Relat
- Conditional → Condition
- Happiness → Happi
- Running → Run

Final Normalized Forms

Word	Final Stem
Relational	Relat
Conditional	Condition
Happiness	Happi
Running	Run

Contribution of Each Technique

- **Derivational Rules:** Identify new word forms.
- **Inflectional Analysis:** Remove grammatical endings.
- **Finite-State Parsing:** Validate word formation.
- **Porter Stemmer:** Produce a common stem.

3. Morphology-Based SEO System

Input Words

- Optimize
- Optimization
- Optimized
- Optimizing

Step 1: Morphological Breakdown

Word	Root	Affix
Optimize	Optimize	—
Optimization	Optimize	-ation
Optimized	Optimize	-ed
Optimizing	Optimize	-ing

Step 2: Classification

- Optimize → Base word
- Optimization → Derivational
- Optimized → Inflectional
- Optimizing → Inflectional

Step 3: Finite-State Parsing

Root → Add suffix

- Optimize → +ation → Optimization
- Optimize → +ed → Optimized
- Optimize → +ing → Optimizing

Step 4: Porter Stemmer

All words are normalized to:

Optimize

Benefits

- Improves keyword indexing.
- Increases search relevance.
- Reduces duplicate words.
- Improves SEO ranking.

4. Hybrid Morphological Pipeline

Input Words

- Compute
- Computer
- Computing
- Computation

Step 1: Morphological Breakdown

Word	Root	Suffix
Compute	Compute	—
Computer	Compute	-er
Computing	Compute	-ing
Computation	Compute	-ation

Step 2: Classification

- Compute → Base word
- Computer → Derivational
- Computing → Inflectional
- Computation → Derivational

Step 3: Finite-State Parsing

Root → Add suffix

- Compute → +er → Computer
- Compute → +ing → Computing
- Compute → +ation → Computation

Step 4: Porter Stemmer

All words are normalized to:

Compute

Final Normalized Form

Word	Final Root
Compute	Compute
Computer	Compute
Computing	Compute
Computation	Compute

5. Morphology-Based Framework

Input Words

- Decide
- Decision
- Decisive
- Deciding

Step 1: Morphological Breakdown

Word	Root	Affix
Decide	Decide	—
Decision	Decide	-sion
Decisive	Decide	-sive
Deciding	Decide	-ing

Step 2: Classification

- Decide → Base word
- Decision → Derivational
- Decisive → Derivational
- Deciding → Inflectional

Step 3: Finite-State Parsing

Root → Add suffix

- Decide → +sion → Decision
- Decide → +sive → Decisive
- Decide → +ing → Deciding

Step 4: Porter Stemmer

All words are normalized to:

Decide

Meaning of Suffixes

- **-sion** → Converts verb into noun.
- **-sive** → Converts verb into adjective.
- **-ing** → Shows present participle.

Final Normalized Form

Word	Final Stem
Decide	Decide
Decision	Decide
Decisive	Decide
Deciding	Decide

Conclusion

A hybrid morphological processing approach combines derivational analysis, inflectional analysis, finite-state parsing, and Porter stemming to identify roots, remove prefixes and suffixes, and normalize different word forms. This improves search accuracy, semantic analysis, document indexing, text mining, and information retrieval by mapping related words to a common base form.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____